

Beyond Gumbel

How the sampling model determines the loss

$$U_j = s_{ij} + \epsilon_j, \epsilon_j \stackrel{\text{i.i.d.}}{\sim} \text{Gumbel}(0, 1), j = 1, \dots, N \longrightarrow \text{sample } \sigma(1)$$

↓

$$P(\sigma(1)=j) = \frac{\exp(s_{ij})}{\sum_k \exp(s_{ik})} \Rightarrow \mathcal{L}_{\text{InfoNCE}} = - \sum_{m=1}^M \log \frac{\exp(s_{ij}^{(m)})}{\sum_k \exp(s_{ik}^{(m)})}$$

$$U_j = \underbrace{s_{ij}}_{\text{score}} + \underbrace{\epsilon_j}_{\text{Gumbel}} \xrightarrow{\text{Gumbel max trick}} P(j^* = j) = \frac{\exp(s_{ij})}{\sum_k \exp(s_{ik})} = \text{softmax}$$

The **additive Gumbel noise** assumption produces the **softmax / cross-entropy** form.

This is *not* a universal law; it is a *modeling choice*.

What happens in the general case?

$$U_j \sim p(u | s_{ij})$$

What is the resulting ranking probability? What loss should we minimize?

$$P(\sigma(1)=j) = ? \qquad \mathcal{L} = ?$$